



# Girijananda Chowdhury University Guwahati 781017, Assam

## Quiz - I (SECTION A)

BCE23405T Structural Analysis

Session: 2024-2025

Batch: 2023-2027

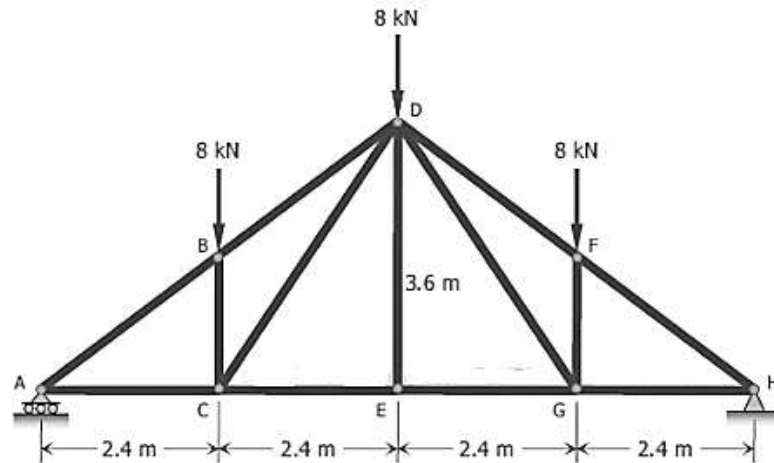
3 February 2026

Total Marks: 10

Duration: 60 min.

### Question 1 (8 marks) [CO1]

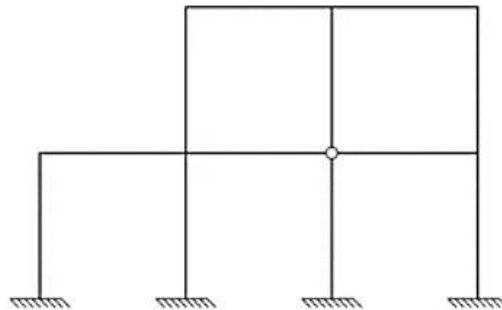
Determine the support reactions and the forces in all the members of the pin-jointed frames



### Question 2 (2 marks)

[CO1]

Determine the degree of indeterminacy and the number of degrees-of-freedom for the pin-jointed and rigid-jointed frames indicated in Figures.





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## Quiz - II (SECTION A)

BCE23405T Structural Analysis

Session: 2024-2025

Batch: 2023-2027

2 April 2026

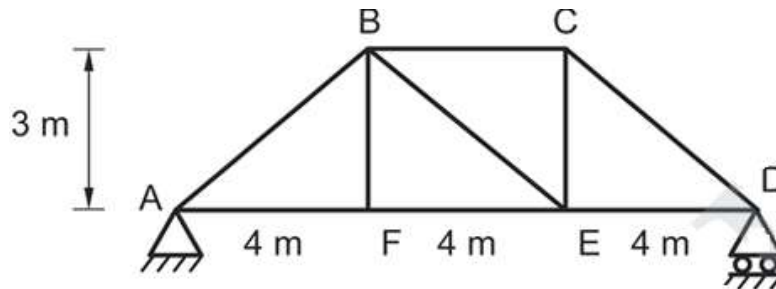
Total Marks: 10

Duration: 60 min.

### Question 1 (4 marks)

[CO2]

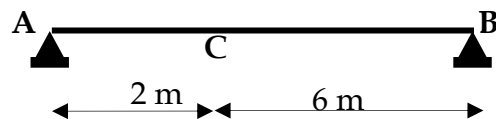
Draw the influence line diagram for members BC, BE, AF and BF of the given truss.



### Question 2 (4 marks)

[CO2]

Draw the influence line diagram for Reactions at A and B, shear and Moment at point C of the given beam



### Question 3 (2 marks)

[CO2]

- Influence lines are especially useful for structures subjected to \_\_\_\_\_ loads.
- The ILD for bending moment in a simply supported beam is \_\_\_\_\_ in shape.



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## Quiz - II (SECTION B)

BCE23405T Structural Analysis

Session: 2024-2025

Batch: 2023-2027

2 April 2026

Total Marks: 10

Duration: 60 min.

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### Question 1 (4 marks)

[CO2]

A hollow steel tube of 200 mm external diameter and 25mm thick is 4m long and used as a column. If one end is fixed and another end is hinged, find the load the column can carry. Use Euler's formula and factor of safety as 2.

### Question 2 (4 marks)

[CO2]

Derive the expression for crippling load for columns

- with both ends hinged (**Roll No: 1 to 9**)
- with both ends fixed (**Roll No: 10 to 19**)
- with one end fixed other hinged (**Roll No: 20 to 29**)
- with one end fixed other free (**Roll No: 30 to 38**)

### Question 3 (2 marks)

[CO2]

- The effective length of a cantilever column is \_\_\_\_\_.
- Rankine's formula combines \_\_\_\_\_ and Euler theory.